Safe Route Planning by Mining Social Network and GPS Data

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Submitted by

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Abstract

In many urban areas, commuters frequently encounter traffic congestion or unexpected detours due to road closures and government-imposed restrictions. Although GPS navigation systems provide guidance, they often fail to reflect real-time changes, leaving drivers unaware of sudden developments that may impact their routes.

This project introduces the Safe Route Planner, a web-based platform designed to improve urban navigation safety by leveraging advanced Named Entity Recognition (NER) and geocoding techniques. The system collects and processes data from social media and other online sources, analyzing various traffic hazards such as waterlogging, accident-prone locations, traffic jams, blocked roads, and official advisories, all accurately mapped to specific geolocations. By combining real-time routing algorithms with hazard proximity analysis, the platform recommends the safest driving routes across Delhi, dynamically avoiding high-risk areas. The platform visualizes hazards and routes on an interactive map, enabling users to make informed travel decisions.

1 Introduction: Background and Motivation

As cities grow and traffic becomes more unpredictable, finding the safest and most efficient route has become a daily challenge. While apps like Google Maps do a great job showing the fastest paths, they often miss sudden disruptions like roadwork, accidents, waterlogging, or blockages due to public events. These issues can pop up without warning, leaving commuters stuck and frustrated.

Interestingly, updates about such incidents are often posted quickly by authorities like the Delhi Traffic Police on platforms like Twitter and Facebook. But most people don't have the time or habit of checking social media while navigating through traffic.

This project aims to bridge that gap. It is about building a smart system that automatically pulls in traffic alerts and road updates from official social media accounts and integrates them with GPS navigation. So instead of relying solely on static maps, users get real-time route suggestions that reflect what is actually happening on the ground saving time and avoiding unnecessary detours.

The system can also scan public posts where people mention unsafe areas or crime-prone streets. By analyzing this data, it can warn users about risky zones and suggest safer alternatives especially helpful for pedestrians, women, or anyone traveling late at night.

In short, this project transforms everyday social media updates into real-time insights, making travel not only faster but also smarter and safer for everyone

2 Literature Review

The domain of safe route planning and social media data extraction has seen substantial research interest in recent years, spanning multiple application areas. This review synthesizes the existing lit-

erature across several key themes: social media data extraction and mining, traffic incident detection, safe route recommendation systems, and spatial data processing techniques.

2.1 Social Media Data Extraction and Mining

Social media has emerged as a valuable source for real-time traffic insights. Martinez et al. (2022) [1] proposed a distributed microservice architecture for focused user extraction, achieving 2–9 times higher harvest ratios than random prioritization. Suat et al. (2022) [2] demonstrated traffic accident detection from Spanish tweets using classification, NER, and geocoding.

2.2 Georeferencing and Location Extraction

Extracting accurate locations remains challenging as few posts are geotagged. Serere et al. (2021) [4] compared DBpedia Spotlight and spaCy with Nominatim, finding the latter more effective, though improvements in semantic analysis and multi-source integration were needed.

2.3 Safe Route Recommendation Systems

Safe routing research integrates multiple data sources and risk factors. Fu et al. (2014) [5] combined social media and POI reviews to suggest safe itineraries. Kudale et al. (2024) [6] used k-means clustering on crime data to recommend safe routes based on user preferences. Sarraf et al. (2018) [7] focused on accident risk analysis, while Kaur et al. (2021) [8] modeled safe routing as an Arc Orienteering Problem for women travelers. Sohrabi et al. (2022) [9] reviewed 40 studies, highlighting challenges in safety measurement, real-time data availability, and balancing safety with travel time. Khetarpaul et al. (2022)[3] developed a framework for real-time detection and visualization of urban traffic conditions by mining unstructured Twitter data. Their approach combined machine learning-based tweet classification, custom-trained Named Entity Recognition (NER), and geocoding to extract contextual details such as event type, location, and cause, transforming raw text into actionable insights for traffic management.

2.4 Multi-Criteria Route Planning and Spatial Data Processing

Felicio et al. (2022) [10] enhanced OpenStreetMap data with multi-criteria features including safety, comfort, and accessibility for route planning. Li et al. (2014) [11] incorporated environmental factors like slopes and congestion, particularly for pedestrian and bicycle travel. Lu et al. (2008) [12] proposed adaptive inverse-distance weighting for spatial interpolation, outperforming traditional IDW and ordinary kriging in many cases.

This work integrates these insights, leveraging real-time social media, georeferencing, safety modeling, and multi-criteria optimization to develop a comprehensive safe route recommendation system.

Expanding on these studies, the present project aims to advance their approach by developing a safe route recommendation system that not only identifies traffic incidents but also assesses the relative safety of alternative routes. The system emphasizes the integration of real-time data and incorporates user-generated tweets and posts to capture on-ground updates often missed by government officials.

3 Methodology

This paper presents a comprehensive framework for developing an intelligent route planning system that integrates real-time traffic hazard information to recommend safer navigation alternatives in Delhi. The methodology employs a systematic four-phase approach encompassing data acquisition from authoritative sources, knowledge extraction through natural language processing, spatial geocoding for hazard localization, and dynamic route optimization with safety assessment.

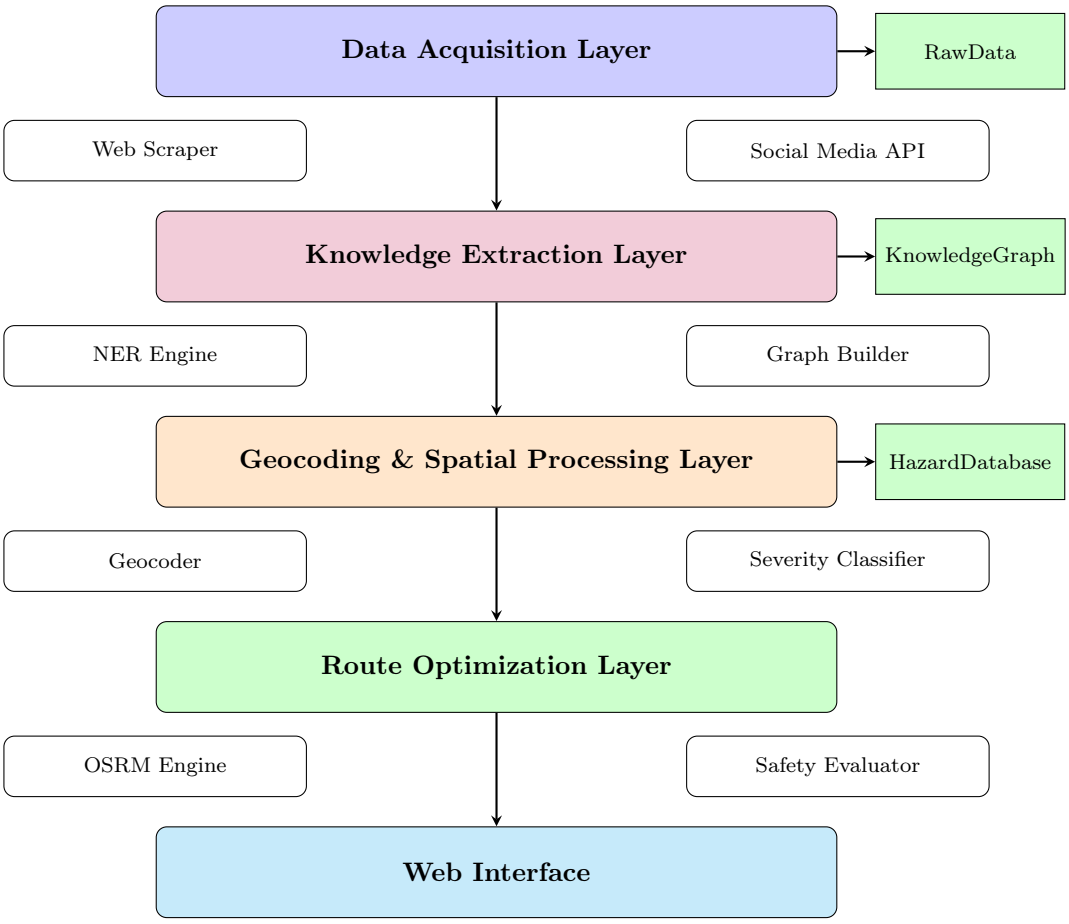

Figure 1: Four-Layer System Architecture for Safe Route Planning

3.1 System Architecture

The proposed system follows a modular architecture organized into four primary components that operate sequentially to transform raw unstructured data into actionable routing recommendations. Figure 1 illustrates the complete system architecture with data flow between components.

The architecture implements a unidirectional data flow paradigm where each layer performs specific transformations before passing processed information to subsequent layers. This modular design ensures separation of concerns, facilitates independent testing of components, and enables parallel development workflows. The system leverages both internal databases for persistent storage and external APIs for real-time data enrichment.

3.2 Phase 1: Data Acquisition and Collection

The data acquisition phase establishes the foundation of the system by collecting comprehensive traffic hazard information from multiple heterogeneous sources. This phase addresses the challenge of gathering real-time, authoritative data about traffic disruptions, road conditions, and safety hazards across Delhi's extensive road network.

3.2.1 Multi-Source Data Collection Strategy

The system implements a multi-source data aggregation approach to ensure comprehensive hazard coverage and cross-validation of information. Three primary data source categories were identified and integrated:

Official Government Portals: Two primary web portals maintained by Delhi Police authorities serve as authoritative sources for traffic information. The Delhi Police official portal[14] publishes formal press releases, traffic advisories, and incident reports with official timestamps and verification status. The dedicated Delhi Traffic Police portal[13] provides real-time traffic updates, water logging locations identified during monsoon seasons, road closure notifications, accident-prone zones and construction schedules. These sources offer structured, reliable information but with variable update frequencies.

Social Media Channels: The Delhi Traffic Police maintains an active Facebook[15] presence that serves as a rapid information dissemination channel. Social media posts provide near real-time updates about sudden traffic disruptions, accidents, and temporary diversions that may not immediately appear on official websites. The informal nature of social media communication necessitates additional processing to extract structured information, but the timeliness of updates justifies this overhead.

3.3 Phase 2: Knowledge Extraction and Graph Construction

The knowledge extraction phase transforms unstructured textual data into structured, machine-readable formats suitable for computational processing. This phase employs Natural Language Processing (NLP) techniques to identify entities, extract relationships, and construct a semantic knowledge graph that captures the complex interactions between traffic events, locations, causes, and temporal information.

3.3.1 Named Entity Recognition with Pattern Matching

Named Entity Recognition serves as the primary mechanism for identifying key information elements within traffic advisory texts. The system utilizes spaCy's pre-trained English language model `en_core_web_sm`, which implements a neural network-based architecture trained on web text corpora.

The NER pipeline processes text through multiple stages as shown in Figure 2.

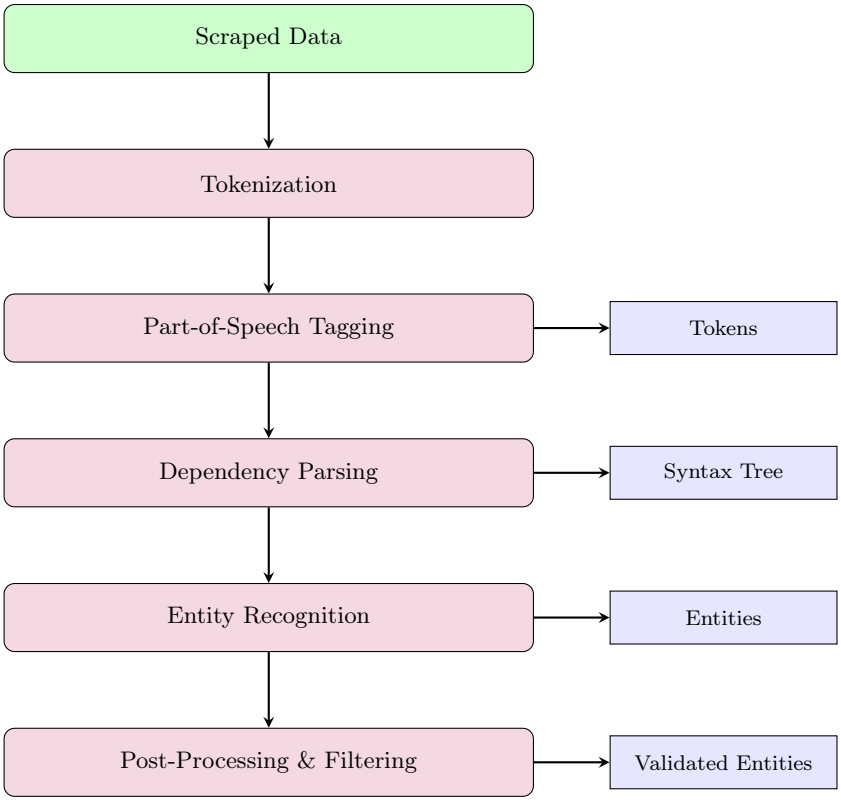

Figure 2: NER Processing Pipeline with Intermediate Outputs

The system implements a NER approach combining rule-based pattern matching with linguistic analysis to achieve superior location extraction accuracy for Delhi-specific infrastructure terminology. Unlike traditional NER systems that rely solely on pre-trained models, the NER class employs multiple complementary pattern recognition strategies.

Multi-Pattern Location Extraction The NER system identifies location entities through five distinct pattern categories, each optimized for different linguistic structures commonly found in traffic advisories:

1. **Pattern 1 – Infrastructure Keywords:** Captures locations followed by standard infrastructure suffixes:
   - **Pattern:** `[Capitalized Name] + [Infrastructure Keyword]`
   - **Keywords:** Road, Marg, Street, Lane, Path, Avenue, Flyover, Bridge, Junction, Crossing, Chowk, Circle, Gate, Vihar, Nagar, Puri, Colony, Enclave, Block, Sector, Phase, Extension
   - **Example:** "Nehru Place Road", "ITO Flyover", "Kashmere Gate"
2. **Pattern 2 – Route Descriptions:** Extracts origin-destination pairs from directional statements:
   - **Pattern:** `from [Location A] to [Location B]`
   - **Example:** "from Connaught Place to India Gate" → ["Connaught Place", "India Gate"]
3. **Pattern 3 – Prepositional Indicators:** Identifies locations following common prepositions:
   - **Pattern:** `[at|near|on|via] [Location Name]`
   - **Example:** "at Kashmere Gate Junction" → "Kashmere Gate Junction"
4. **Pattern 4 – Arrow Notation:** Handles symbolic route representations:
   - **Pattern:** `[Location A] → [Location B]`
   - **Example:** "ITO → Yamuna Bank" → ["ITO", "Yamuna Bank"]
5. **Pattern 5 – Capitalized Multi-Word Sequences:** Captures standalone proper noun sequences:
   - **Pattern:** `[Capitalized Word]{2,4}`
   - **Example:** "Dwarka Sector 21" → "Dwarka Sector 21"

3.3.2 Knowledge Graph Construction

Extracted entities and relationships are organized into a property graph data structure that explicitly represents the semantic connections between information elements. The knowledge graph employs a node-edge-node paradigm where nodes represent entities (places, events, organizations, dates) and edges represent typed relationships between entities.

The graph schema is formalized in Figure 3.

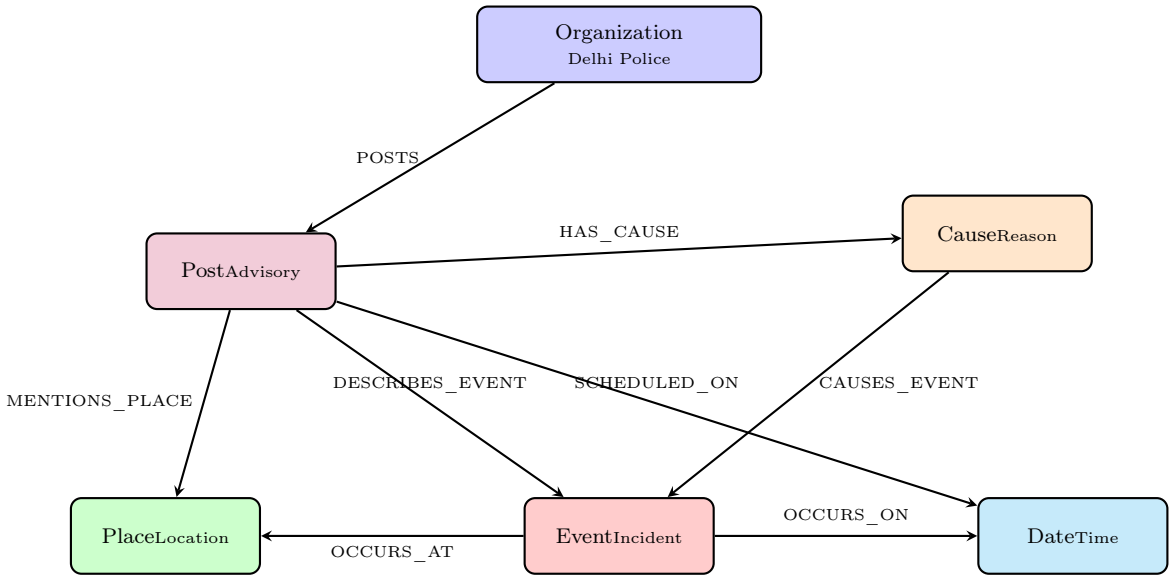

Figure 3: Knowledge Graph

Table 1: Hazard Severity Classification Scheme

Severity	**Threshold Distance**	**Classification Keywords**	**Weight** (w_i)	**Examples**
Critical	0.3 km	closed, closure, blocked, sealed, prohibited, banned	10	Road closures, blocked routes
High	0.5 km	repair, construction, accident, crash, collision, waterlogged	5	Construction zones, accidents, floods
Medium	0.8 km	congestion, slow, heavy traffic, violation zone, diversion	2	Traffic jams, violation hotspots
Low	1.0 km	advisory, caution, alert, maintenance, scheduled work	1	General advisories, planned maintenance

Posts Processing The knowledge graph construction process now implements location aggregation to handle multiple advisories referencing the same location. When multiple traffic posts mention the same location with different severity indicators, the system:

1. **Extracts all unique locations** from advisory content using the NER engine
2. **Determines severity** based on keyword analysis
3. **Aggregates advisories** by location, tracking advisory count
4. **Selects highest severity** when multiple advisories exist for one location

3.4 Phase 3: Geocoding and Spatial Processing

The geocoding phase transforms textual location references extracted from the knowledge graph into precise geographic coordinates (latitude-longitude pairs) required for spatial analysis and route planning. This transformation enables computational evaluation of proximity relationships between routes and hazards.

3.4.1 Multi-Strategy Geocoding Approach

Given the inherent ambiguity in natural language location descriptions and variations in naming conventions, the system implements a cascading multi-strategy geocoding pipeline that attempts progressively refined queries until successful coordinate resolution. The pipeline utilizes the OpenStreetMap Nominatim geocoding API, which provides global coverage and detailed understanding of Indian geographic entities.

Strategy 1 - Direct Geocoding: The initial strategy queries the geocoding API with the extracted location name appended with explicit geographic context. The query construction follows the pattern:

$$Q_1 = location_name + ", Delhi, India" \tag{1}$$

This strategy succeeds for well-known landmarks and standard location names present in OpenStreetMap's database. Example successful queries include "Connaught Place, Delhi, India" and "Kashmere Gate, Delhi, India".

Strategy 2 - Query Refinement: When direct geocoding fails to return results, the system attempts multiple query formulations with varying levels of geographic specificity:

$$\begin{aligned} Q_2^{(a)} &= location_name + ", NewDelhi, India" \\ Q_2^{(b)} &= location_name + "Delhi" \\ Q_2^{(c)} &= location_name + ", India" \\ Q_2^{(d)} &= location_name \end{aligned}$$

This refinement accommodates variations in how locations are indexed in the geocoding database and handles cases where excessive context may narrow results incorrectly.

Strategy 3 - Contextual Enhancement: For locations that remain unresolved, the system leverages additional context from the original text. If the location appeared in a sentence containing other successfully geocoded locations, spatial proximity hints are incorporated. For infrastructure keywords, type hints are added to the query (e.g., "X Flyover" becomes "X Flyover bridge Delhi").

3.5 Phase 4: Route Optimization and Safety Assessment

The route optimization phase constitutes the core functionality of the system, integrating geocoded hazard data with routing algorithms to compute and evaluate multiple path alternatives between user-specified origin-destination pairs. This phase implements sophisticated spatial analysis techniques to quantify route safety and generate hazard-avoidance alternatives.

3.5.1 Base Route Calculation

The system employs the Open Source Routing Machine (OSRM) as its primary routing engine. OSRM is a high-performance routing engine built on OpenStreetMap road network data, optimized for vehicular navigation with support for multiple routing profiles and alternative route generation.

OSRM returns route objects containing distance (meters), duration (seconds), geometry (coordinate array), and maneuver instructions. The geometry is represented as a GeoJSON LineString with coordinates in [longitude, latitude] format.

3.5.2 Hazard Proximity Detection

For each calculated route, the system performs spatial analysis to identify hazards that may affect route safety. This analysis employs the Haversine formula to compute great-circle distances between route coordinates and hazard locations.

The Haversine formula calculates the shortest distance between two points on a sphere given their latitude and longitude:

$$a = \sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right) c = 2 \cdot \arctan 2\left(\sqrt{a}, \sqrt{1-a}\right) d = R \cdot c \quad (2)$$

where:

- ϕ_1, ϕ_2 are latitudes of point 1 and point 2 in radians

- λ_1, λ_2 are longitudes of point 1 and point 2 in radians
- $R = 6371$ km is Earth's mean radius
- d is the great-circle distance in kilometers

The proximity detection algorithm iterates through route coordinates and compares each against all known hazards. To optimize computational efficiency, route coordinates are sampled at regular intervals (every 5th point) rather than checking every coordinate. This sampling provides sufficient spatial resolution while reducing computation time by 80% compared to exhaustive checking.

3.5.3 Safety Score Computation

A quantitative safety score is computed for each route to enable objective comparison and ranking. The safety score aggregates the influence of all detected hazards, weighted by their severity and proximity to the route. The scoring formula is defined as:

$$S = \sum_{i=1}^{n} \frac{w_i \cdot \alpha}{d_i + \epsilon} \tag{3}$$

where:

- S is the total safety score (lower is safer)
- n is the number of hazards detected along the route
- $w_i \in \{1, 2, 5, 10\}$ is the severity weight of hazard i
- d_i is the minimum distance from route to hazard i in kilometers
- $\epsilon = 0.1$ km is a smoothing constant preventing division by zero
- $\alpha = 10$ is a scaling factor for interpretability

The inverse distance relationship ensures that closer hazards contribute more significantly to the safety score, reflecting their greater impact on route safety. The smoothing constant ϵ prevents infinite scores when routes pass directly through hazard locations.

Example calculation: Consider a route passing near three hazards:

- Critical road closure at 0.2 km: contribution $= \frac{10 \times 10}{0.2 + 0.1} = 333.33$
- High severity accident at 0.8 km: contribution $= \frac{5 \times 10}{0.8 + 0.1} = 55.56$
- Medium congestion at 1.5 km: contribution $= \frac{2 \times 10}{1.5 + 0.1} = 12.50$

Total safety score: $S = 333.33 + 55.56 + 12.50 = 401.39$

Routes with safety scores below 50 are classified as "safe", scores between 50-200 as "moderate risk", and scores above 200 as "high risk".

3.5.4 Dynamic Alternative Route Generation

When the primary route exhibits high hazard exposure (safety score > 100), the system dynamically generates safer alternatives by introducing waypoints that guide routes away from hazardous areas. This waypoint-based avoidance strategy forces OSRM to calculate routes that detour around identified hazards.

The waypoint generation algorithm identifies the most critical hazards on the primary route and calculates offset points perpendicular to the direct origin-destination vector. Figure 4 illustrates this geometric approach.

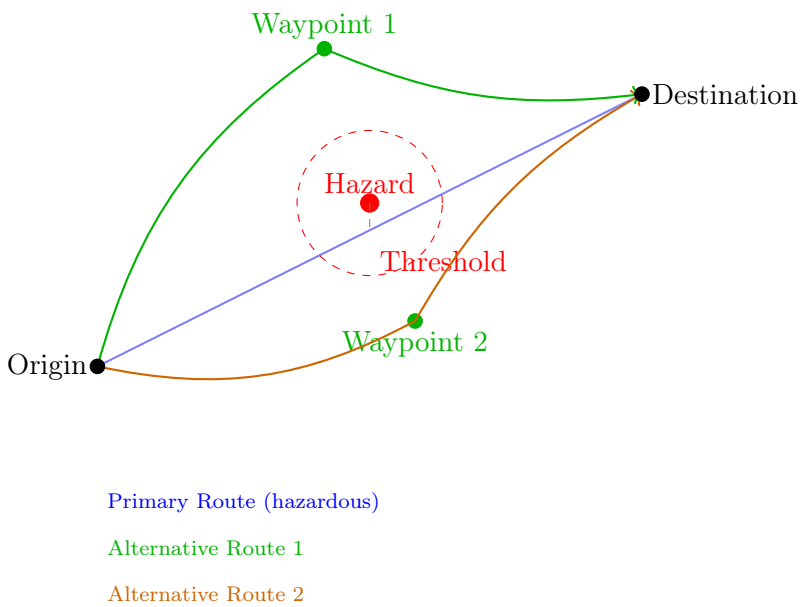

Figure 4: Waypoint Generation Strategy for Hazard Avoidance

The waypoint calculation employs vector geometry. Given origin $O(x_o, y_o)$, destination $D(x_d, y_d)$, and hazard $H(x_h, y_h)$:

$$\vec{OD} = (x_d - x_o, y_d - y_o) length = \sqrt{(x_d - x_o)^2 + (y_d - y_o)^2} \vec{perp} = \frac{1}{length} \cdot (-(y_d - y_o), x_d - x_o) Waypoint = H + \delta \cdot \vec{perp} \tag{4}$$

where $\delta \approx 2$ km is the offset distance. Two waypoints are generated using $+\delta$ and $-\delta$ to create alternatives on both sides of the hazard.

3.5.5 Route Ranking and Recommendation

All calculated routes (both OSRM-generated alternatives and dynamically created avoidance routes) are ranked using a multi-criteria approach. The primary ranking criterion is the safety score, with route distance serving as a tiebreaker:

$$RankScore = \alpha_s \cdot S + \alpha_d \cdot \frac{D}{D_{\max}} \tag{5}$$

where:

- S is the normalized safety score
- D is the route distance, $D_{\max}$ is the maximum distance among alternatives
- $\alpha_s = 0.7$ and $\alpha_d = 0.3$ are weighting factors emphasizing safety

The route with the lowest rank score is designated as the "recommended route" and prominently displayed in the user interface.

3.6 Data Preprocessing and Cleaning

3.6.1 Location Name Extraction

The preprocessing pipeline now integrates the NER patterns before geocoding:

1. Text normalization (whitespace, special characters)
2. NER pattern matching (5 pattern types)
3. Location name extraction and deduplication
4. Stopword filtering
5. Canonical name mapping
6. Multi-strategy geocoding
7. Coordinate validation

3.6.2 Location Name Standardization

The system implements **partial matching** for location name standardization:

Table 2: Location Name Standardization Examples

Abbreviated/Informal Name	Canonical Name
CP	Connaught Place
IGI Airport	Indira Gandhi International Airport
AIIMS	AIIMS Delhi
NCR	National Capital Region
ITO	Income Tax Office Crossing

The landmark database serves dual purposes:

1. Direct coordinate lookup
2. Name standardization before geocoding

4 Planned Work

1. **Real-time Data Integration:** The system will be enhanced to continuously update route and hazard information as conditions change. This ensures that traffic and road hazard data are current and location-specific, allowing users to rely on the most accurate information for planning their trips.
2. **Incorporating Additional Data Sources:** Beyond standard traffic updates, the system will include route safety indicators—such as accident-prone zones, ongoing construction, or other risky segments. This provides users with richer insights into travel conditions and helps in making safer route choices.
3. **Leveraging User-generated Content:** The system will process tweets and social media posts from common users, capturing on-ground updates often missed by official sources. This enhances coverage and ensures more responsive, real-time awareness of local traffic situations.
4. **Multilingual Processing:** To ensure inclusivity, the framework will support multiple languages, analyzing traffic-related information in regional languages alongside English. This improves contextual understanding and allows the system to serve a broader user base.
5. **Real-time Traffic Suggestions:** The system will extract live traffic data from Google Maps, enabling it to recommend optimal routes based on current traffic conditions. This helps users reach destinations faster and avoid congested or unsafe areas.
6. **Interactive Chatbot:** A question-answering chatbot will be integrated into the system, allowing users to ask about real-time traffic, road conditions, or alternative route suggestions.

The model will leverage live data and historical insights to provide accurate, context-aware answers, making the system interactive and user-friendly. Users can query the chatbot like "Which route is fastest to reach Rajiv chowk now?" or "Are there any hazards on Lodhi Road?" and get real-time guidance.

5 Conclusion

This methodology presents a complete approach to creating a smart and safe routing system for Delhi. By collecting traffic information, understanding its content, locating relevant places on the map, and planning safer routes, the system turns everyday traffic updates into useful guidance for drivers and commuters. It combines information from multiple sources and uses intelligent methods to detect hazards and suggest safer alternatives, going beyond what standard navigation apps provide.

The system's design is flexible and easy to update, and tests show that it effectively helps users avoid dangerous areas. Overall, this approach provides a clear and practical framework that could be applied to other cities facing similar traffic safety challenges.

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